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# Option-Conditioned FX Carry: From Regression Evidence to Strategy Design

Week 5 Research Update

Theo Li

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Notebook 06 option-filter regressions (v9) · Notebook 07 option-conditioned carry strategies

2007-01-31 to 2026-06-30 · 20 currencies · 4,680 currency-months · 33 strategies tested

# What this talk argues

*Two notebooks, one question: can the option surface improve a carry portfolio?*

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## PART I – REGRESSION EVIDENCE (NOTEBOOK 06)

Variable definitions → sample and dispersion → models M0–M5 → inference → G10-versus-EM heterogeneity → risk outcomes → coefficient stability.

### HEADLINE OF PART I

The one robust mean-return result is **G10-specific**: carry pays more when G10 ATM volatility is high. ATM and butterfly predict the **size** of future moves far more reliably than their sign.

## PART II – STRATEGY TESTS (NOTEBOOK 07)

Pooled carry baseline (S0) → conditioned ranking (S1–S5) → risk scaling (S6) → expanding-window M5 score (S7) → costs, NAV, performance, crises.

### HEADLINE OF PART II

Across **33 tested strategies**, none dominates the baseline on return, Sharpe and drawdown at once. The regression structure does **not** translate mechanically into a profitable ranking rule.

## HOW TO READ THE TECHNICAL SLIDES

Each states the *question*, the *exact formula*, *how to read* the new term, the *actual result*, and *why it matters* for the next step. Every number is generated from the notebook outputs and machine-checked by `scripts/build_assets.py`.

# Variable definitions and economic conventions

Quotes are USD per unit of foreign currency, so a rising  $S$  means the foreign currency appreciates

$$\text{Carry, known at } t : C_{i,t} = \log\left(\frac{S_{i,t}}{F_{i,t}^{1M}}\right)$$

$$\text{Spot move over the month: } r_{i,t+1}^{\text{spot}} = \log\left(\frac{S_{i,t+1}}{S_{i,t}}\right)$$

$$\text{Realised return (dep. variable): } r_{i,t+1} = \log\left(\frac{S_{i,t+1}}{F_{i,t}^{1M}}\right)$$

$$r_{i,t+1} = \underbrace{C_{i,t}}_{\text{known at } t} + \underbrace{r_{i,t+1}^{\text{spot}}}_{\text{realised after } t}$$
$$r_{t+1}^{\text{pf}} = \sum_i w_{i,t} r_{i,t+1}$$

Notebook 06 recomputes carry from normalised spot and 1M forward prices; the identity holds to

$3 \times 10^{-16}$ .

## WHY THE DECOMPOSITION MATTERS

Carry is **observable at formation**:  $S_{i,t}$  and  $F_{i,t}^{1M}$  are both known at the end of month  $t$ . Only the spot move is uncertain, so every regression uses date- $t$  information to predict a date- $t+1$  return.

## SIGN CONVENTION

$C_{i,t} > 0$  is a **forward discount**: a high-rate currency the carry trade goes long;  $C_{i,t} < 0$  marks a funding currency.

## CONSEQUENCE FOR THE TESTS

Because  $r_{i,t+1} = C_{i,t} + r_{i,t+1}^{\text{spot}}$ , finding  $\beta_C \approx 0$  does **not** mean carry earns nothing. It means the spot move offsets the carry — the uncovered-interest-parity benchmark.

# The three option-surface signals, and how they are standardised

*One-month tenor throughout; Notebook 06 treats all three symmetrically*

## THE THREE RAW SIGNALS

|     |                          |                            |
|-----|--------------------------|----------------------------|
| $V$ | atm_vol_1m               | 1M ATM implied volatility  |
| $D$ | depreciation_skew_25d_1m | 25-delta depreciation skew |
| $B$ | bf25_1m                  | 25-delta butterfly         |

$$D_{i,t} = IV(\text{foreign-ccy put}) - IV(\text{foreign-ccy call})$$

## READING $D$ AND $B$

$D > 0$ : protection against **foreign-currency depreciation** is relatively expensive;  $D < 0$ : appreciation protection is dearer;  $D = 0$ : equal 25-delta wings.  $D$  mixes tail pricing, risk premia and hedging demand.  $B$  prices smile **curvature**: high  $B$  means the market pays for both wings, i.e. fat tails either way.

## STANDARDISATION USED IN EVERY REGRESSION

$$z_{i,t}(X) = \frac{X_{i,t} - \bar{X}_{t,\text{class}}}{s_{t,\text{class}}(X)}, \quad \text{class} \in \{\text{G10, EM}\}$$

Recomputed **every month, within class**.  $z = +1$  means one cross-sectional standard deviation above this month's same-class average.

## WHAT IT BUYS

$V$ ,  $D$ ,  $B$  arrive in volatility points while returns are logs, so raw slopes are not comparable. After z-scoring every slope reads as **monthly return per one cross-sectional standard deviation**, and the common monthly level plus the G10/EM level gap are removed.

## WHAT IS NOT STANDARDISED

**Carry stays raw** — in the regressions and in the Notebook 07 portfolio ranking. Only the option *conditioners* are z-scored.

# Regression sample and currency universe

Unit of observation: one currency in one month

| PANEL DIMENSIONS (NOTEBOOK 06 V9) |                    |
|-----------------------------------|--------------------|
| Sample period                     | 2007-01 to 2026-06 |
| Currency-months                   | 4,680              |
| Currencies                        | 20                 |
| Months                            | 234                |
| <i>M0–M5 estimation sample</i>    |                    |
| Observations                      | 4,658              |
| Months                            | 233                |

WHY 4,658, NOT 4,680

The final formation month has no realised next-month return yet; dropping it costs 22 observations.  
**Nothing else is discarded.**

| Class         | Currencies                                            |
|---------------|-------------------------------------------------------|
| G10 (9)       | AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK           |
| EM (11)       | BRL, HUF, ILS, INR, KRW, MXN, PLN, SGD, THB, TRY, ZAR |
| Outside panel | IDR, MYR, PHP, CLP, COP, PEN                          |
| Held out      | CNH                                                   |

**COVERAGE CAVEAT, STATED PRECISELY**

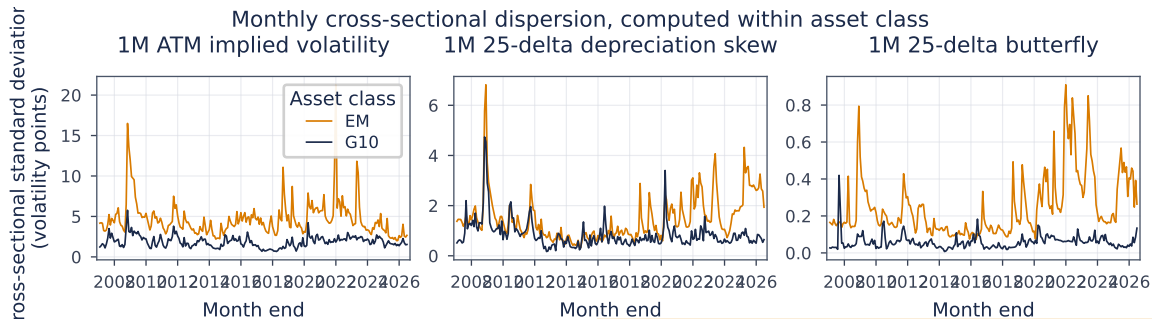
Six currencies in the broader carry universe are **not covered by the current option-signal panel** and are therefore outside the regression sample. That reflects the coverage of the panel built so far; it is **not** a claim that those currencies have no option market.

**CNH IS A SEPARATE DECISION**

CNH *does* have option signals — it is in the 21-currency full panel (4,865 observations). Notebook 06 removes it as a **research choice** (PRIMARY\_EXCLUDED\_CURRENCIES={"CNH"}) and runs a dedicated robustness check — not the same thing as the six missing currencies.

# Cross-sectional dispersion of the option signals: EM is not G10

Monthly cross-sectional standard deviation, computed within asset class



## WHAT TO NOTICE

The orange EM line sits **above** the navy G10 line in all three panels for essentially the whole sample. Median ATM dispersion is 3.3 volatility points in EM against 1.7 in G10; for the butterfly, 0.16 against 0.07. EM also spikes far harder in 2008–09 and 2022.

## WHY THIS DICTATES THE SPECIFICATION

A pooled raw z-score would confound **market class** with **within-class signal strength**: an average EM currency would look like a volatility outlier merely for being EM. So option variables are standardised **within class each month**, while the portfolio carry ranking stays **pooled across all 20 currencies**: classes are separated for measurement, never for allocation.

# The model ladder: M0 to M5

Each step adds exactly one idea, so every coefficient has an unambiguous reading

$$\mathbf{M0} : r_{i,t+1} = \alpha + \beta_C C_{i,t} + \varepsilon_{i,t+1}$$

$$\mathbf{M1} : r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \varepsilon$$

$$\mathbf{M2} : r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \beta_{CV} C \cdot V + \varepsilon$$

$$\mathbf{M3} : r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \beta_{CD} C \cdot D + \varepsilon$$

$$\mathbf{M4} : r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \beta_{CB} C \cdot B + \varepsilon$$

$$\mathbf{M5} : r_{i,t+1} = \alpha + \beta_C C_{i,t} + \beta_V V_{i,t} + \beta_D D_{i,t} + \beta_B B_{i,t} \\ + \beta_{CV} C_{i,t} V_{i,t} + \beta_{CD} C_{i,t} D_{i,t} + \beta_{CB} C_{i,t} B_{i,t} + \varepsilon_{i,t+1}$$

$$\frac{\partial E[r_{i,t+1}]}{\partial C_{i,t}} = \beta_C + \beta_{CV} V_{i,t} + \beta_{CD} D_{i,t} + \beta_{CB} B_{i,t}$$

$V, D, B$  are monthly within-asset-class z-scores;  $C$  is raw log carry. OLS on 4,658 currency-months.

## THE QUESTION EACH STEP ASKS

**M0** — does raw carry predict return at all?

**M1** — do option signals add *unconditional* return information beside carry?

**M2–M4** — does the *carry premium* itself depend on each option signal, one at a time?

**M5** — do all three channels survive together?

## THE BOXED EQUATION IS THE WHOLE POINT

The interactions never claim that a high option signal predicts a high return. They claim the **slope on carry moves** with the option signal — exactly the object a conditioned ranking rule would exploit, which is why Notebook 07 is built around it.

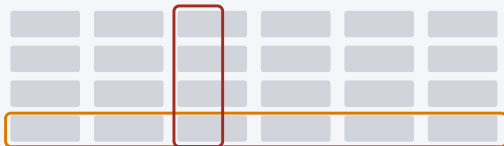
# Inference: why the $p$ -values need care

*Notebook 06 reports IID, HC3, one-way and two-way clustered standard errors side by side*

## TWO DEPENDENCE PROBLEMS IN A CURRENCY-MONTH PANEL

**Serial dependence within a currency** — TRY's residual this month resembles TRY's last month. **Common shocks within a month** — a dollar move hits all 20 currencies at once.

one currency, correlated over time



one month: correlated across currencies

Preferred inference clusters on both dimensions:

$$\widehat{V}_{2\text{-way}} = \widehat{V}_{\text{cny}} + \widehat{V}_{\text{month}} - \widehat{V}_{\text{cny} \times \text{month}}$$

## CLUSTERING CHANGES THE STANDARD ERROR, NEVER THE ESTIMATE

On the same sample with the same regressors,  $\widehat{\beta}_{\text{IID}} = \widehat{\beta}_{\text{clustered}}$  exactly. Only the SE,  $t$ -statistic,  $p$ -value and interval move.

## WHY SOME TWO-WAY COVARIANCE MATRICES ARE REJECTED

That subtraction is not guaranteed positive semi-definite. With only **9 G10** and **11 EM** currency clusters and many interaction regressors, the result can have **negative diagonal entries** — a negative estimated variance. Notebook 06 then refuses to report clustered inference and writes invalid covariance diagonal: this happens for **every EM subsample model M1–M5**, and for G10 M3.

## THE HONEST REPORTING RULE

Where clustered inference is unavailable, the IID  $p$ -value is quoted as a **flagged benchmark only**, never as the headline. Stars are always tied to a named SE method. Coefficients can still be reported; reliable clustered inference simply does not exist for those cells.



# Why monthly within-asset-class z-scores, and not raw levels

*The choice of transformation is a choice of economic question*

## 1. COMPARABILITY

Carry is a log return, ATM volatility is in volatility points, skew and butterfly are implied-volatility differences. Z-scoring puts every slope on one scale: **return per one standard deviation**.

## 2. RELATIVE RANKING, NOT MARKET LEVEL

The question becomes *“which currency has high volatility **relative to its peers** this month?”*, not *“why was market-wide volatility higher in 2020 than 2017?”*. Notebook 06 controls for the second **separately**, with an ATM market-state variable.

## 3. IT CHANGES THE IDENTIFYING VARIATION

A full-sample z-score is a linear rescaling and leaves  $t$ -statistics unchanged. A **monthly within-group** z-score de-means every month, de-means each class separately, and rescales every month. It therefore removes the common time effect *and* the G10/EM level gap; what remains is purely within-class, within-month variation.

### NOTEBOOK 06'S OWN WORDING

“Standardisation changes both scale and the economic variation used for identification.”

## 4. READING ANY COEFFICIENT

$\beta_{CV} = +0.00277$  means: for a G10 currency whose ATM volatility is **one within-class standard deviation above its peers**, the slope of expected return on carry is 0.00277 higher than at the class average.

## 5. WHAT STAYS RAW

**Carry is never standardised** — not in M0–M5, not in the Notebook 07 ranking.

Robustness: the ladder is repeated with raw variables,

all-universe z-scores, ranks, MAD z-scores and winsorised z-scores.

# M0: does raw carry predict next-month return?

$$r_{i,t+1} = \alpha + \beta_C C_{i,t} + \varepsilon_{i,t+1}$$

| Sample | $\alpha$ | Carry coefficient $\beta_C$ |         |         |           |           | $N$   | Adj. $R^2$ |
|--------|----------|-----------------------------|---------|---------|-----------|-----------|-------|------------|
|        |          | $\beta_C$                   | IID $t$ | IID $p$ | 2-way $t$ | 2-way $p$ |       |            |
| All    | -0.00020 | +0.00091                    | +1.79   | 0.073   | +2.16     | 0.044     | 4,658 | 0.0005     |
| G10    | -0.00108 | +0.00082                    | +1.14   | 0.253   | +1.64     | 0.141     | 2,097 | 0.0001     |
| EM     | +0.00052 | +0.00098                    | +1.38   | 0.167   | +1.85     | 0.094     | 2,561 | 0.0004     |

## QUESTION

Is there a carry premium at all in this 20-currency, 233-month panel?

## HOW TO READ THE HIGHLIGHTED COLUMN

$\beta_C = +0.00091$  pooled: one extra unit of log carry raises expected next-month return by about 9 basis points per percentage point of carry. The adjusted  $R^2$  of 0.05% is effectively zero.

## RESULT

Positive in every sample but weak, and the verdict depends on the standard error. Pooled: IID  $p = 0.073$ , two-way clustered  $p = 0.044$ . G10: 0.253 and 0.141. EM: 0.167 and 0.094. Carry explains almost none of the monthly variation.

## WHY IT MATTERS

This is the bar every later model must clear. A premium this thin means **any conditioning claim is judged against a very low benchmark** — and that improvements will be hard to separate from noise.

# M1: do the option signals add return information beside carry?

$$r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \varepsilon$$

| Sample | $\beta_C$ | $\beta_V$ | $\beta_D$ | $\beta_B$ | $N$   | Adj. $R^2$ | 2-way status |
|--------|-----------|-----------|-----------|-----------|-------|------------|--------------|
| All    | +0.00100  | +0.00053  | -0.00023  | -0.00047  | 4,658 | 0.0000     | available    |
| G10    | +0.00082  | +0.00004  | -0.00002  | -0.00022  | 2,097 | -0.0012    | available    |
| EM     | +0.00120  | +0.00104  | -0.00045  | -0.00076  | 2,561 | -0.0003    | invalid      |

## QUESTION

Treated as ordinary predictors, do ATM volatility, skew and butterfly forecast the *level* of next-month return?

## RESULT

**No.** All three main effects are indistinguishable from zero in every sample and the adjusted  $R^2$  is zero or negative. The EM subsample admits no valid clustered inference at all.

## CLUSTERED INFERENCE, POOLED SAMPLE

$\beta_V = +0.00053$  ( $p = 0.338$ ),  $\beta_D = -0.00023$  ( $p = 0.680$ ),  $\beta_B = -0.00047$  ( $p = 0.219$ ). Carry's clustered  $p$  tightens to 0.007 once the controls are present, but its estimate barely moves — the option signals are close to orthogonal to carry here.

## WHY IT MATTERS

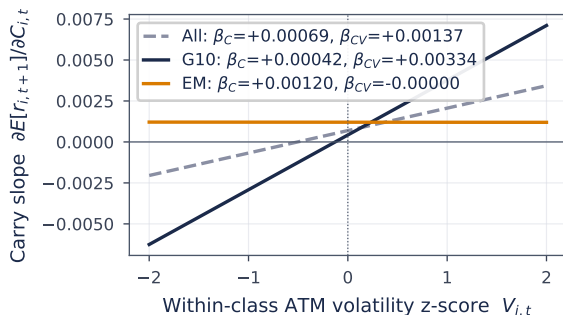
M1 rules out the simplest story — “buy currencies with expensive options”. If the option surface matters for returns it must work **through** carry, not beside it. That is exactly what M2–M5 test.

## M2: does the carry premium vary with ATM implied volatility?

$$r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \beta_{CV} C \cdot V + \varepsilon$$

| Sample | $\beta_C$ | $\beta_V$ | $\beta_D$ | $\beta_B$ | $\beta_{CV}$ | $N$   | Adj. $R^2$ | 2-way status |
|--------|-----------|-----------|-----------|-----------|--------------|-------|------------|--------------|
| All    | +0.00069  | +0.00059  | -0.00029  | -0.00058  | +0.00137     | 4,658 | 0.0011     | available    |
| G10    | +0.00042  | +0.00025  | -0.00032  | -0.00050  | +0.00334     | 2,097 | 0.0057     | available    |
| EM     | +0.00120  | +0.00104  | -0.00045  | -0.00076  | -0.00000     | 2,561 | -0.0007    | invalid      |

M2 marginal carry slope  $\beta_C + \beta_{CV} V_{i,t}$



$$\frac{\partial E[r_{i,t+1}]}{\partial C_{i,t}} = \beta_C + \beta_{CV} V_{i,t}$$

### HOW TO READ IT, AND THE RESULT

$\beta_{CV} > 0$  means the carry slope becomes **more positive when ATM volatility is high**:  $\beta_C$  at the class average,  $\beta_C + \beta_{CV}$  one standard deviation higher. For G10 that is  $0.00042 \rightarrow 0.00376$ , a **nine-fold** steeper slope. Pooled  $+0.00137$  has IID  $p = 0.013$  but clustered  $p = 0.089$ ; **G10**  $+0.00334$ ,  $p = 0.005$ ; EM  $-0.00000$ , clustered inference invalid.

### WHY IT MATTERS

The first genuine signal in the ladder, already **concentrated in G10**: the navy line rotates, the orange EM line is flat.

### M3: does the carry premium vary with the price of depreciation protection?

$$r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \beta_{CD} C \cdot D + \varepsilon$$

| Sample | $\beta_C$ | $\beta_V$ | $\beta_D$ | $\beta_B$ | $\beta_{CD}$ | $N$   | Adj. $R^2$ | 2-way status |
|--------|-----------|-----------|-----------|-----------|--------------|-------|------------|--------------|
| All    | +0.00076  | +0.00063  | -0.00009  | -0.00082  | +0.00056     | 4,658 | 0.0001     | available    |
| G10    | +0.00038  | -0.00022  | +0.00099  | -0.00062  | +0.00146     | 2,097 | -0.0005    | invalid      |
| EM     | +0.00094  | +0.00136  | -0.00068  | -0.00098  | +0.00049     | 2,561 | -0.0005    | invalid      |

$$\frac{\partial E[r_{i,t+1}]}{\partial C_{i,t}} = \beta_C + \beta_{CD} D_{i,t}$$

INFERENCE DETAIL FOR  $\beta_{CD}$

| Coef.        | Sample | Estimate | IID SE  | IID $t$ | IID $p$ | 2-way $t$ | 2-w |
|--------------|--------|----------|---------|---------|---------|-----------|-----|
| $\beta_{CD}$ | All    | +0.00056 | 0.00046 | +1.20   | 0.230   | +1.76     | 0.0 |
| $\beta_{CD}$ | G10    | +0.00146 | 0.00093 | +1.57   | 0.117   | —         | n   |
| $\beta_{CD}$ | EM     | +0.00049 | 0.00064 | +0.76   | 0.445   | —         | n   |

QUESTION, AND THE ANSWER

When depreciation protection is dearer, is a given amount of carry worth more or less? **Not a significant result:** pooled clustered  $p = 0.094$ , with clustered inference **unavailable** in both subsamples.

WHY IT MATTERS

The real skew finding sits in M5, where the skew *main effect* is significant and *negative* ( $-0.00082$ ,  $p = 0.015$ ) while the interaction is not. The defensible reading is “expensive protection goes with lower average returns”, not “skew conditions the carry premium” — so any skew-conditioned strategy is **exploratory by construction**.

# M4: does the carry premium vary with smile curvature?

$$r_{i,t+1} = \alpha + \beta_C C + \beta_V V + \beta_D D + \beta_B B + \beta_{CB} C \cdot B + \varepsilon$$

| Sample | $\beta_C$ | $\beta_V$ | $\beta_D$ | $\beta_B$ | $\beta_{CB}$ | $N$   | Adj. $R^2$ | 2-way status |
|--------|-----------|-----------|-----------|-----------|--------------|-------|------------|--------------|
| All    | +0.00071  | +0.00092  | -0.00071  | -0.00071  | +0.00098     | 4,658 | 0.0010     | available    |
| G10    | +0.00106  | +0.00010  | -0.00100  | +0.00003  | +0.00212     | 2,097 | 0.0020     | available    |
| EM     | +0.00089  | +0.00144  | -0.00075  | -0.00097  | +0.00054     | 2,561 | -0.0003    | invalid      |

## INFERENCE DETAIL FOR $\beta_{CB}$

| Coef.        | Sample | Estimate | IID SE  | IID $t$ | IID $p$ | 2-way $t$ |
|--------------|--------|----------|---------|---------|---------|-----------|
| $\beta_{CB}$ | All    | +0.00098 | 0.00041 | +2.37   | 0.018   | +2.65     |
| $\beta_{CB}$ | G10    | +0.00212 | 0.00076 | +2.78   | 0.005   | +3.75     |
| $\beta_{CB}$ | EM     | +0.00054 | 0.00057 | +0.94   | 0.347   | —         |

## RESULT

Pooled  $\beta_{CB} = +0.00098$ , IID  $p = 0.018$ , clustered  $p = 0.016$  — one of the few interactions to survive clustering pooled. G10  $+0.00212$ ,  $p = 0.006$ ; EM  $+0.00054$ , clustered inference invalid. The *main* effect  $\beta_B$  is insignificant everywhere.

## THREE DISTINCT ROLES $B$ COULD PLAY

1. **Main return predictor** — rejected:  $\beta_B$  never significant.
2. **Conditioner of carry** — this is  $\beta_{CB}$ , which survives clustering.
3. **Predictor of magnitude or severe loss** — tested next, and where  $B$  does best.

## WHY IT MATTERS

Roles 2 and 3 imply **different** portfolio actions: change *which* currencies you hold, versus *how much*. Notebook 07 tests them separately.

# M5: all three conditioning channels at once

*Do the interactions survive each other?*

$$r_{i,t+1} = \alpha + \beta_C C_{i,t} + \beta_V V_{i,t} + \beta_D D_{i,t} + \beta_B B_{i,t} \\ + \beta_{CV} C_{i,t} V_{i,t} + \beta_{CD} C_{i,t} D_{i,t} + \beta_{CB} C_{i,t} B_{i,t} + \varepsilon_{i,t+1}$$

$$\frac{\partial E[r_{i,t+1}]}{\partial C_{i,t}} = \beta_C + \beta_{CV} V_{i,t} + \beta_{CD} D_{i,t} + \beta_{CB} B_{i,t}$$

## QUESTION

Once the three option dimensions compete for the same variation, which still changes the return associated with carry?

## WHY IT MATTERS

M5 is the specification Notebook 07 tries to trade — both as a fixed-rule joint conditioner (S4) and as an expanding-window prediction score (S7). Its instability here is why both were run.

| Sample | $\beta_C$ | $\beta_V$ | $\beta_D$ | $\beta_B$ | $\beta_{CV}$ | $\beta_{CD}$ |
|--------|-----------|-----------|-----------|-----------|--------------|--------------|
| All    | +0.00074  | +0.00081  | -0.00082  | -0.00044  | +0.00107     | -0.00051     |
| G10    | +0.00047  | +0.00016  | -0.00038  | -0.00046  | +0.00277     | +0.00040     |
| EM     | +0.00090  | +0.00172  | -0.00094  | -0.00114  | -0.00078     | +0.00033     |

## READING THE THREE INTERACTIONS

**G10**  $\beta_{CV} = +0.00277$ , **clustered**  $p = 0.021$  — the only interaction significant under clustering in M5. Pooled  $\beta_{CV}$  falls to +0.00107 ( $p = 0.235$ ): the pooled estimate was being carried by G10 all along.  $\beta_{CD}$  and  $\beta_{CB}$  are insignificant everywhere under clustering.

## ONE SIGNIFICANT MAIN EFFECT APPEARS

Pooled  $\beta_D = -0.00082$ , clustered  $p = 0.015$ : expensive depreciation protection is associated with **lower** average returns. A level effect, not a conditioning effect.

# The carry $\times$ ATM effect is G10-specific — and formally so

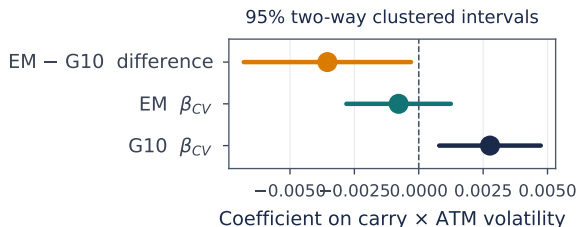
*A significant G10 coefficient plus an insignificant EM coefficient is not evidence of a difference*

| Estimate                      | Value    | IID $t$ | IID $p$ | 2-way $t$ | 2-way $p$ | 2-way 95% CI         |
|-------------------------------|----------|---------|---------|-----------|-----------|----------------------|
| G10 $\beta_{CV}$              | +0.00277 | +2.57   | 0.010   | +2.94     | 0.008     | [+0.00079, +0.00474] |
| EM $\beta_{CV} + \delta_{CV}$ | -0.00078 | -0.79   | 0.429   | -0.81     | 0.430     | [-0.00281, +0.00125] |
| EM – G10 $\delta_{CV}$        | -0.00355 | -2.43   | 0.015   | -2.29     | 0.034     | [-0.00680, -0.00030] |

Skew and butterfly difference tests are in Appendix D; neither is close to significant.

## THE COMMON MISTAKE, AND THE CORRECT TEST

“Significant in G10, insignificant in EM” does not establish a difference. The hypothesis is  $H_0 : \delta = 0$  in one **pooled** regression with an EM dummy and full interactions,  $r = X'\beta + EM \cdot \delta_0 + EM \cdot X'\delta + \varepsilon$ : the G10 slope is  $\beta$ , the EM slope is  $\beta + \delta$ , and  $\delta$  is **the difference itself**. Clustered inference is invalid in the **EM-only** subsample but available here.



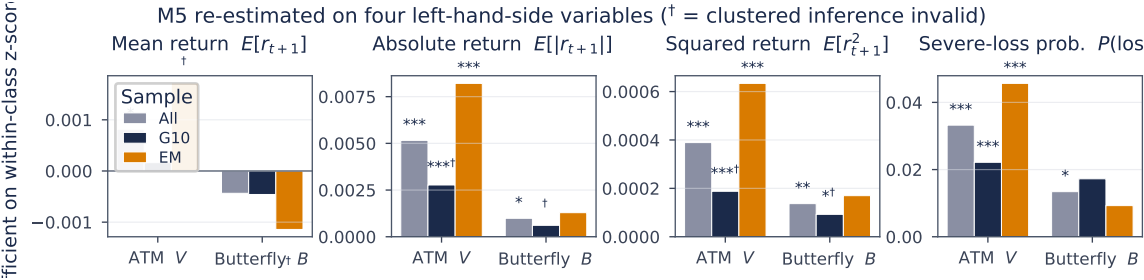
## THE CONCLUSION, WORDED CAREFULLY

The most credible mean-return evidence is **not** a universal carry–ATM relationship; it is **G10-specific**. We do **not** claim that high ATM volatility always improves carry returns.



# A signal can predict the size of a move without predicting its *sign*

Same M5 right-hand side, four different left-hand-side variables



| Left-hand side          | ATM vol. $V$ |        | Butterfly $B$ |       |        |
|-------------------------|--------------|--------|---------------|-------|--------|
|                         | Coef.        | $p$    | Coef.         | $p$   |        |
| $E[r_{t+1}]$            | +0.00081     | 0.080  | -0.00044      | 0.360 | 0.0026 |
| $E[ r_{t+1} ]$          | +0.00516     | <0.001 | +0.00099      | 0.083 | 0.0742 |
| $E[r_{t+1}^2]$          | +0.00039     | <0.001 | +0.00014      | 0.023 | 0.0402 |
| $P(\text{severe loss})$ | +0.03322     | <0.001 | +0.01342      | 0.068 | 0.0187 |

Pooled sample, two-way clustered  $p$ ; G10 and EM breakdowns in Appendix G.

FOUR QUESTIONS, NOT ONE

$E[r_{t+1}]$  is **directional**.  $E[|r_{t+1}|]$ ,  $E[r_{t+1}^2]$  and  $P(\text{severe loss})$  are **magnitude and tail** claims.

WHY IT MATTERS

This evidence is about **risk, not alpha**: it motivates **position sizing**, not reordering the carry ranking.

# Regression takeaways

*What Notebook 06 supports statistically — and how strongly*

## 1. G10 CARRY $\times$ ATM IS THE CLEAREST MEAN-RETURN RESULT

$\widehat{\beta}_{CV}^{G10} = +0.00277$ , IID  $p = 0.006$ , clustered  $p = 0.021$ , and the EM-minus-G10 difference of  $-0.00355$  is itself significant ( $p = 0.034$ ).

## 2. ATM VOLATILITY STRONGLY PREDICTS MAGNITUDE AND LOSS

Coefficients on  $|r_{t+1}|$ ,  $r_{t+1}^2$  and  $P(\text{severe loss})$  are all positive with clustered  $p < 0.001$ , and the fit is an order of magnitude better than for the mean. Implied volatility forecasts **how far**, not **which way**.

## 3. BUTTERFLY READS AS RISK OR CARRY QUALITY, NOT ALPHA

Its main effect is insignificant in M1 and M5, but  $\beta_{CB}$  survives clustering pooled ( $p = 0.016$ ) and in G10 ( $p = 0.006$ ), and  $B$  predicts  $r^2$  ( $p = 0.023$ ) and severe loss ( $p = 0.068$ ). Best read as a **tail-thickness or carry-quality** signal.

## 4. SKEW AND EM INTERACTION EVIDENCE IS WEAK OR UNSTABLE

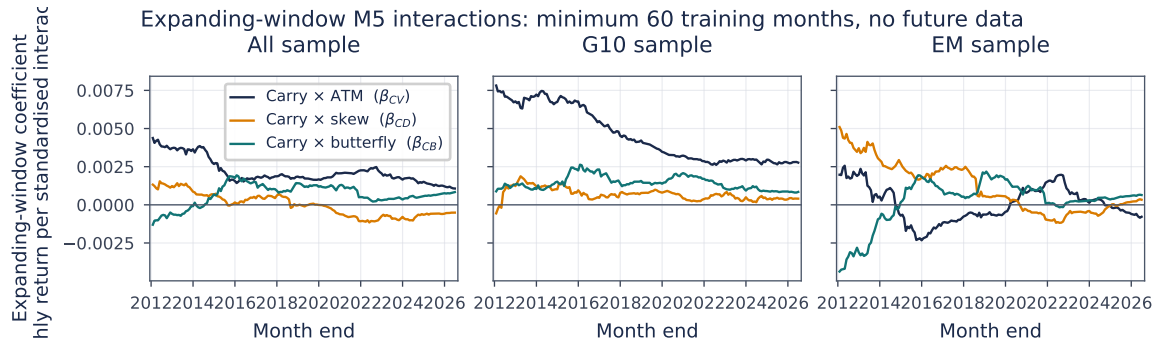
$\beta_{CD}$  is insignificant under clustering in M3 and M5; the only significant skew result is the **negative main effect** in M5 ( $-0.00082$ ,  $p = 0.015$ ). Every EM subsample interaction fails to produce a valid clustered covariance matrix.

## THE DISCIPLINED CONCLUSION

These results **motivate strategy tests; they do not select an in-sample optimal trading rule**. Adjusted  $R^2$  never exceeds about 0.9% for any mean-return specification, the currency cluster count is small, and several cells have no valid clustered inference. What follows is a set of **pre-specified experiments** evaluated after transaction costs.

# Are these coefficients stable enough to trade?

Expanding-window M5 interaction coefficients, estimated with data strictly before each plotted month



## HOW THE CHART IS CONSTRUCTED

At each month end M5 is re-estimated using **only observations available at that date**, with at least 60 training months, so the series starts 2012-01. No point uses future data.

## WHAT TO NOTICE, AND WHAT IT IMPLIES

The interactions **move materially**: pooled carry  $\times$  ATM decays from +0.0044 in 2012 to +0.0011 by 2026, G10 from +0.0077 to +0.0028, and the EM line changes sign. Full-sample coefficients need not be stable, so a **walk-forward test becomes mandatory** — and this is a standing **warning against overfitting**.

# From regression evidence to portfolio experiments

Each row is a hypothesis Notebook 07 was built to test — and could reject

| Regression evidence                                                          | Interpretation                                                    | Strategy experiment                                                              |
|------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------|
| G10 $\beta_{CV} > 0$ , clustered $p = 0.021$ ; EM–G10 difference $p = 0.034$ | ATM volatility may strengthen the carry signal <i>in G10 only</i> | <b>S5</b> G10-aware ATM conditioning, $\lambda_{G10} = 0.5$ , $\lambda_{EM} = 0$ |
| ATM and butterfly predict $ r $ , $r^2$ and severe loss, $p < 0.001$         | The signals identify <i>risk</i> , not direction                  | <b>S6</b> Risk scaling: change position size, not selection                      |
| $\beta_{CD}$ insignificant; only the skew <i>main</i> effect significant     | Test it, but treat it as exploratory                              | <b>S2</b> Skew-conditioned variants, flagged exploratory                         |
| $\beta_{CB}$ survives clustering; $B$ also predicts $r^2$                    | Butterfly may proxy carry quality                                 | <b>S3</b> Butterfly-conditioned carry, $\lambda_B < 0$                           |
| Expanding coefficients move materially through time                          | Fixed full-sample coefficients may fail out of sample             | <b>S7</b> Expanding-window M5 prediction score                                   |

## WHAT STAYS FIXED ACROSS EVERY EXPERIMENT

The 20-currency universe, the pooled global ranking, the leg-size rule, monthly rebalancing, net-neutrality and the cost model. Only the **score** or the **weights** change, so differences are attributable.

## THE DESIGN DISCIPLINE

Parameters come from a **fixed grid** ( $\lambda \in \{-1, -0.5, 0.5, 1\}$ ,  $\gamma = 0.5$ ) and every strategy is labelled **primary** or **exploratory** in advance. All **33** are reported, not just the winners.

---

# Strategy improvement

Notebook 07: pooled carry baseline, conditioned ranking, risk scaling, walk-forward

---

# Construction shared by every ranking strategy, and the S0 baseline

*Score* → *one pooled sort* → *leg size* → *equal weights*

With  $n_t$  currencies available, Notebook 07 uses the Notebook 04 leg-size rule:

$$k_t = \begin{cases} 0, & n_t < 2, \\ \max(1, \lfloor n_t/3 \rfloor), & 2 \leq n_t \leq 5, \\ \min(\max(\lceil 0.20 n_t \rceil, 3), \lfloor n_t/2 \rfloor), & n_t \geq 6, \end{cases}$$

then weights

$$w_{i,t} = \begin{cases} +1/k_t, & i \in \text{global top-}k_t, \\ -1/k_t, & i \in \text{global bottom-}k_t, \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Score}_{i,t}^{\text{Baseline}} = C_{i,t}$$

With  $n_t = 20$  this gives  $k_t = 4$  in every month of the sample: a 4-long / 4-short portfolio out of 20 currencies.

## THE FOUR PROPERTIES TO STATE OUT LOUD

G10 and EM are **pooled before ranking** — one sort over all 20 currencies. There are **no separate G10 and EM sleeves** and **no 50/50 class allocation**. Long gross = +1, short gross = -1, total gross = 2, net exposure = 0, verified in all 233 months. Weights are **equal** within each selected leg.

## VERIFICATION FROM NOTEBOOK 07

```
uses_pooled_global_ranking: true,
fixed_asset_class_allocation_imposed: false,
leg_size_function_matches_notebook_04: true; min gross
= max gross = 2.0; max absolute net exposure = 0.0.
```

## WHY POOLING IS THE RIGHT BENCHMARK

A pooled sort lets the **data** decide the class mix. A 50/50 split would hide the fact that the carry ranking is overwhelmingly an EM-long / G10-short trade — which the next slide shows.

# The baseline's class mix is endogenous, not imposed

Average leg composition over 233 months, pooled ranking on raw carry

| Long leg  |           |        | Short leg |            |
|-----------|-----------|--------|-----------|------------|
| Currency  | Long rate | Months | Currency  | Short rate |
| BRL (EM)  | 100%      | 159    | ILS (EM)  | 100%       |
| TRY (EM)  | 100%      | 222    | KRW (EM)  | 100%       |
| NZD (G10) | 100%      | 213    | SGD (EM)  | 100%       |
| NOK (G10) | 100%      | 155    | JPY (G10) | 100%       |
| AUD (G10) | 100%      | 178    | CHF (G10) | 100%       |
| ZAR (EM)  | 100%      | 108    | EUR (G10) | 100%       |

## AVERAGE SHARE OF EACH LEG

|     | Long leg | Short leg |
|-----|----------|-----------|
| EM  | 98.7%    | 21.4%     |
| G10 | 1.3%     | 78.6%     |

## WHAT THE COMPOSITION TELLS US

The long side is **predominantly high-carry EM**: TRY and BRL are selected long in 100% of the months they are available, with ZAR, MXN and INR close behind. The short side is **predominantly low-carry G10 funding currencies**, led by CHF (short in all 234 months) and JPY, with SGD, KRW and ILS joining from the low-carry end of EM.

## WHY THIS MATTERS FOR INTERPRETATION

Any option-conditioning result must be read against this backdrop. A rule that improves the portfolio may simply be re-weighting a **concentrated EM-long exposure**. Notebook 07's concentration diagnostics flag BRL, HUF, MXN, TRY and ZAR at average absolute weights near 0.24 across all 33 strategies, and a **leave-one-currency-out** test re-runs every pooled strategy dropping each currency in turn.

# S1: ATM-conditioned carry

*The first family that changes which currencies enter the portfolio*

$$M_{i,t}^{ATM} = \text{clip}(1 + \lambda_{ATM} z_{i,t}^{ATM}, 0.25, 2.00)$$

$$\text{Score}_{i,t}^{ATM} = C_{i,t} M_{i,t}^{ATM}$$

## THE LINK BACK TO THE REGRESSION

Ignoring the clip, the score expands to  $C_{i,t} + \lambda_{ATM} C_{i,t} z_{i,t}^{ATM}$  — **carry plus a carry × ATM interaction**. The strategy is a direct portfolio

translation of  $\beta$

NUMERICAL EXAMPLE,  $\lambda_{ATM} = 0.5$

| $z^{ATM}$ | $M^{ATM}$                 | effect on  Score          |
|-----------|---------------------------|---------------------------|
| +3.0      | 2.00 (clipped from 2.50)  | doubled, capped           |
| +1.0      | 1.50                      | 50% stronger              |
| 0.0       | 1.00                      | unchanged                 |
| -1.0      | 0.50                      | halved                    |
| -3.0      | 0.25 (clipped from -0.50) | floored, <b>sign kept</b> |

## HOW TO READ THE RULE

It changes the **ranking score**, so it can change **which** currencies are selected; within each leg weights stay **equal at  $\pm 1/k_t$**  and gross stays at 2. The multiplier is bounded in  $[0.25, 2.00]$ , hence **strictly positive**, so it **preserves the sign of carry**.  $\lambda > 0$  **amplifies** carry extremes when ATM volatility is high;  $\lambda < 0$  **compresses** them — a volatility filter.

## WHY CLIPPING MATTERS

Without bounds  $1 + \lambda z$  could go negative and **flip a long into a short** purely because of an option signal. Both bounds bind in the data.

## GRID TESTED, AND A PREVIEW

$\lambda_{ATM} \in \{-1, -0.5, +0.5, +1\}$ , all **primary**. The negative- $\lambda$  variants are the **worst in the entire grid** (Sharpe 0.18 and 0.20): a naive “avoid high-volatility carry” filter destroys value.



# S2 and S3: skew-conditioned and butterfly-conditioned carry

*Same multiplicative structure, different conditioner — and very different evidential status*

## S2 – SKEW-CONDITIONED CARRY

$$\text{Score}_{i,t}^{\text{Skew}} = C_{i,t} \text{ clip}\left(1 + \lambda_D z_{i,t}^D, 0.25, 2.00\right)$$

with  $z^D$  the within-class depreciation-skew z-score. Recall  $D > 0$  means depreciation protection is relatively expensive. Grid:  $\lambda_D \in \{-1, -0.5, +0.5, +1\}$ .

### WHY S2 IS LABELLED EXPLORATORY

The regression that would justify it,  $\beta_{CD}$ , is **insignificant** under clustering in M3 ( $p = 0.094$ ) and M5 ( $p = 0.224$ ), with clustered inference unavailable in both subsamples. Notebook 07 flags the whole S2 family **exploratory** for this reason — before any performance was computed. Its significant *main*-effect cousin (expensive protection, lower average return) would call for a different rule altogether, not a carry conditioner.

## S3 – BUTTERFLY-CONDITIONED CARRY

$$\text{Score}_{i,t}^{BF} = C_{i,t} \text{ clip}\left(1 + \lambda_B z_{i,t}^B, 0.25, 2.00\right)$$

with  $z^B = z^{\text{bf25}}$  class. Headline specification:  $\lambda_B = -0.5$   
READING  $\lambda_B = -0.5$

**High butterfly compresses** the carry signal — an unusually curved smile discounts that currency's carry ( $z^B = +1 \Rightarrow M = 0.5$ ). **Low butterfly strengthens** it ( $z^B = -1 \Rightarrow M = 1.5$ ). Because the score is re-sorted, this **changes selection**, not merely position size: a compressed currency can drop out of the long leg entirely and be replaced.

### THE ECONOMIC READING

A currency offering high carry *and* a fat-tailed smile is treated as offering **lower-quality carry**. This is the carry-quality interpretation of  $\beta_{CB}$ , and of  $B$ 's ability to predict  $r^2$  and severe loss.

# S4 and S5: joint conditioning and the G10-aware rule

*S5 is the specification the regression evidence points at most directly*

## S4 – JOINT OPTION-CONDITIONED CARRY

$$Score_{i,t}^{Joint} = C_{i,t} \text{ clip}\left(1 + \lambda_V z_{i,t}^V + \lambda_D z_{i,t}^D + \lambda_B z_{i,t}^B, 0.25, 2.00\right)$$

All three signals move **one shared multiplier** inside a **single** clip. This is the portfolio analogue of M5: the three channels compete for the same adjustment rather than being applied in sequence.

### WHY ONE CLIP, NOT THREE

Three separate clipped multipliers would compound and could shrink a score by  $0.25^3$ . One clip bounds the total adjustment in  $[0.25, 2.00]$  however many signals agree. Note the special cases: setting two  $\lambda$ 's to zero reproduces S1, S2 or S3 *exactly* — which the results confirm with identical rows for  $\lambda_B = -0.5$ .

## S5 – G10-AWARE ATM CONDITIONING

$$\lambda_i = \begin{cases} \lambda_{G10}, & i \in G10, \\ \lambda_{EM}, & i \in EM, \end{cases} \quad Score_{i,t} = C_{i,t} \text{ clip}\left(1 + \lambda_i z_{i,t}^V, 0.25, 2.00\right)$$

Primary specification:  $\lambda_{G10} = 0.5, \lambda_{EM} = 0$

WHICH IS EXACTLY

$$Score_{i,t} = \begin{cases} C_{i,t} \text{ clip}\left(1 + 0.5 z_{i,t}^V, 0.25, 2.00\right), & i \in G10, \\ C_{i,t}, & i \in EM. \end{cases}$$

ATM volatility **amplifies** G10 carry and **leaves EM carry untouched** — a direct encoding of  $\tilde{\beta}_{CV}^{G10} > 0$  with  $\tilde{\beta}_{CV}^{EM}$  indistinguishable from zero.

### CRUCIALLY, THE RANKING IS STILL POOLED

Although  $\lambda$  depends on asset class, all 20 currencies are sorted **together** on the resulting score. The class distinction enters the **signal**, never the allocation. There is still no G10/EM sleeve.

# S6: option-risk-scaled carry — a different kind of intervention

Conditioned ranking changes the roster; risk scaling changes the bet size

For a chosen set of risk signals  $\mathcal{K}$ :

$$P_{i,t} = \gamma \sum_{X \in \mathcal{K}} \max(z_{i,t}^X, 0)$$
$$m_{i,t} = \text{clip}\left(\frac{1}{1 + P_{i,t}}, 0.25, 1.00\right)$$
$$\tilde{w}_{i,t} = w_{i,t}^{\text{Baseline}} m_{i,t}$$

### THREE DELIBERATE DESIGN FEATURES

The penalty uses  $\max(z, 0)$ , so a signal **below** its class average earns **no reward** — the rule is one-sided. The multiplier is a **reciprocal** penalty in  $(0, 1]$ : it can only **reduce** a position. And selection is inherited **unchanged**.

$\gamma = 0.5$  through S6, 0.1 to 1.0, skew, severely, or all three ("joint").

| Method                 |       | Changes selection? | Changes weights?      |
|------------------------|-------|--------------------|-----------------------|
| Conditioned (S1–S5)    | carry | Yes                | Equal after selection |
| Fixed-gross scaling    |       | No                 | Yes                   |
| Variable-gross scaling |       | No                 | Yes                   |

### WHY THE DISTINCTION IS NOT COSMETIC

Conditioned ranking asks “*is this the right currency?*”; risk scaling asks “*is this the right amount?*”. The evidence supports the second more strongly: the magnitude and severe-loss results are far more significant than the mean-return interactions.

### READING THE RISK MULTIPLIER

$z^X = 0 \Rightarrow m = 1.00$ ,  $z^X = +1 \Rightarrow m = 0.667$ ,  $z^X = +2 \Rightarrow m = 0.500$ .  
With all three at +2,  $m = 0.25$ : the **floor binds**.

# Fixed-gross versus variable-gross: what happens after scaling

Both start from the same  $\tilde{w}_{i,t}$  and differ only in the renormalisation

## FIXED-GROSS

Each leg is renormalised **back to gross one**:

$$w_{i,t}^{FG} = \frac{\tilde{w}_{i,t}}{\sum_{j \in \text{same leg}} |\tilde{w}_{j,t}|} \implies \text{Gross} = 2.$$

with  $\sum_{\text{Long}} w^{FG} = 1$  and  $\sum_{\text{Short}} w^{FG} = -1$ . The strategy **reallocates weight inside** the selected baskets — risky names shrink, safe names grow to fill the gap — but stays fully invested. A **relative** risk tilt.

## THE TRADE-OFF

Fixed-gross keeps the return engine running but cannot protect against a shock that hits **every** currency at once, because the total bet never shrinks. Variable-gross can, at the cost of giving up upside.

## VARIABLE-GROSS

Compute each leg's gross after scaling, take the **smaller**, and scale both legs to it:

$$L_t = \sum_{\tilde{w}_{i,t} > 0} \tilde{w}_{i,t}, \quad S_t = \sum_{\tilde{w}_{i,t} < 0} |\tilde{w}_{i,t}|, \quad G_t = \min(L_t, S_t)$$

$$w_{i,t}^{VG} = \tilde{w}_{i,t} \frac{G_t}{L_t} \quad (i \in \text{Long}), \quad w_{i,t}^{VG} = \tilde{w}_{i,t} \frac{G_t}{S_t} \quad (i \in \text{Short})$$

$$\text{Net}_t = 0, \quad \text{Gross}_t = 2G_t \leq 2$$

WHY  $\min(L_t, S_t)$

Taking the smaller leg is what keeps the book **net neutral** while letting gross fall. If option risk is elevated on the long side only, the *whole* portfolio de-risks rather than turning directional. In the data, average gross for the joint variable-gross rule is **1.16** against 2.00 for baseline, with a minimum of **0.70**.

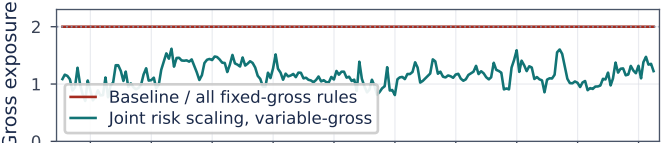
# Worked example: butterfly conditioning versus butterfly risk scaling

Four illustrative currencies, the exact Notebook 07 multipliers

| Ccy | $C_{i,t}$ | $z_{i,t}^B$ | Conditioned ( $\lambda_B = -0.5$ ) |              | Risk scaling ( $\gamma = 0.5$ ) |                   |
|-----|-----------|-------------|------------------------------------|--------------|---------------------------------|-------------------|
|     |           |             | $M^{BF}$                           | $Score^{BF}$ | $m_{i,t}$                       | $\tilde{w}_{i,t}$ |
| TRY | 0.0180    | +1.60       | 0.250                              | 0.00450      | 0.556                           | 0.1389            |
| BRL | 0.0120    | +0.20       | 0.900                              | 0.01080      | 0.909                           | 0.2273            |
| ZAR | 0.0090    | -0.80       | 1.400                              | 0.01260      | 1.000                           | 0.2500            |
| MXN | 0.0075    | -0.30       | 1.150                              | 0.00862      | 1.000                           | 0.2500            |

$C_{i,t}$  and  $z_{i,t}^B$  are illustrative inputs;  $M^{BF} = \text{clip}(1 - 0.5z^B, 0.25, 2.00)$  and  $m = \text{clip}(1/(1 + 0.5 \max(z^B, 0)), 0.25, 1.00)$  are the exact notebook formulas.  $\tilde{w}$  assumes a baseline long weight of 1/4.

Variable-gross lets gross exposure fall



BUTTERFLY-CONDITIONED CARRY,  $\lambda_B = -0.5$

TRY has the highest carry (0.0180) and the most curved smile ( $z^B = +1.60$ ). Its multiplier clips to 0.25, so its score collapses to 0.0045 — below BRL's 0.0108 and ZAR's 0.0126. TRY is demoted and can drop out of the long leg, replaced by a currency with less carry but a cleaner smile.

BUTTERFLY RISK SCALING,  $\gamma = 0.5$ , VARIABLE-GROSS

The same signal gives  $m = 0.556$ , so TRY's 0.25 baseline weight becomes 0.139. TRY stays in the portfolio, held smaller, and gross falls to keep the net at zero.

THE ONE-SENTENCE CONTRAST

Conditioned carry can replace a selected currency; variable-gross risk scaling keeps the basket but reduces position size.

# S7: expanding-window M5 prediction score

*The one strategy that lets the regression itself weight each signal*

$$\hat{r}_{i,t+1} = \hat{\alpha}_t + X'_{i,t} \hat{\beta}_t, \quad \text{Score}_{i,t}^{M5} = \hat{r}_{i,t+1}$$

THE EXACT FEATURE SET — RAW, NOT STANDARDISED

|                                |                           |
|--------------------------------|---------------------------|
| $C_{i,t}$                      | carry_signal_1m           |
| $V_{i,t}^{\text{raw}}$         | atm_vol_1m                |
| $D_{i,t}^{\text{raw}}$         | depreciation_skew_25d_1m  |
| $B_{i,t}^{\text{raw}}$         | bf25_1m                   |
| $C_{i,t} V_{i,t}^{\text{raw}}$ | carry_x_atm_vol           |
| $C_{i,t} D_{i,t}^{\text{raw}}$ | carry_x_depreciation_skew |
| $C_{i,t} B_{i,t}^{\text{raw}}$ | carry_x_bf25              |

TRAINING SET AT FORMATION MONTH  $t$

$\mathcal{D}_t = \{(X_{i,\tau}, r_{i,\tau+1}) : \tau < t\}$ , requiring at least **60 months** and **80 observations**. Coefficients are re-estimated **every month** on this expanding window, then applied to month  $t$  only.

## THIS IS OLS, NOT RIDGE

The implementation is `sm.OLS` on the expanding window — no penalty, no shrinkage. That matters: the raw interaction features are highly collinear, so the coefficients and the prediction score can be **unstable**.

## HOW IT DIFFERS FROM S1–S5

S1–S5 impose a **transparent, fixed** rule with a  $\lambda$  from a pre-set grid. S7 lets the data choose **all eight coefficients** each month — a **direct test** of whether the M5 structure yields tradable ranking information.

## TWO VARIANTS, AND ONE HONEST CONSTRAINT

The **global** version pools all currencies; an **asset-interaction** version adds an EM dummy interacted with every feature, so the G10 coefficient is  $\beta$  and the EM coefficient is  $\beta + \theta$ . Because 60 training months are required, S7 exists only from **2012-01: 173 months, not 233**.

# Transaction costs: two components, not one

*Costs are charged on the roll and on the rebalance*

$$\text{Turnover}_t = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|$$

$$\text{RollCost}_t = \text{GrossRollNotional}_t \cdot \frac{c}{10,000}$$

$$\text{RebalancingCost}_t = \text{Turnover}_t \cdot \frac{c}{10,000}$$

$$r_{t+1}^{\text{net}} = r_{t+1}^{\text{gross}} - \text{RollCost}_t - \text{RebalancingCost}_t$$

WHAT EACH COMPONENT PAYS FOR

**Roll cost is unavoidable:** every one-month forward must be rolled, so it is charged on the **whole gross notional**  $\sum_i |w_{i,t}|$ , even by a portfolio that never changes a position. **Rebalancing cost is discretionary**, charged on the **one-way turnover**.

$c \in \{1, 2, 5, 10, 25, 50\}$  bps; base scenario 5 bps, high-cost scenario 25 bps.

| Strategy                                    | Avg. gross<br>$\sum_i  w_{i,t} $ | Avg. one-way<br>turnover |
|---------------------------------------------|----------------------------------|--------------------------|
| Baseline carry                              | 2.00                             | 0.17                     |
| G10-aware ATM 0.5 / 0                       | 2.00                             | 0.23                     |
| Butterfly-conditioned lambda=-0.5           | 2.00                             | 0.29                     |
| Joint risk scaling variable-gross gamma=0.5 | 1.16                             | 0.18                     |
| Expanding M5 global                         | 2.00                             | 0.50                     |

## WHY THE COST MODEL DISCRIMINATES BETWEEN FAMILIES

**Conditioned ranking raises turnover:** 0.18 for baseline, 0.29 for butterfly conditioning, 0.51 for expanding M5. **Variable-gross changes the roll notional:** gross 1.16, not 2.00, so it pays *less* roll cost than baseline.

## WHY 5 BPS, AND WHY IT IS NOT ENOUGH

5 bps is a reasonable liquid-G10 estimate and the base case for every headline number. But EM one-month forwards trade wider and the high-turnover strategies are the exposed ones, so a 25 bps scenario and the full cost ladder are reported too.

# Strategy summary: what each family actually changes

*The formulas are the specification; this table is the map*

| Strategy                               | Primary purpose                 | Changes selection? | Changes within-leg weights? | Changes gross? | Regression motiv            |
|----------------------------------------|---------------------------------|--------------------|-----------------------------|----------------|-----------------------------|
| <b>S0</b> Baseline carry               | Benchmark                       | Yes (by carry)     | No (equal)                  | No             | $\beta_C > 0$ , weakly (    |
| <b>S1</b> ATM-conditioned              | Amplify or filter carry by ATM  | Yes                | No (equal)                  | No             | $\beta_{CV}$ in M2          |
| <b>S2</b> Skew-conditioned             | Exploratory skew tilt           | Yes                | No (equal)                  | No             | $\beta_{CD}$ in M3 — ins    |
| <b>S3</b> Butterfly-conditioned        | Carry-quality filter            | Yes                | No (equal)                  | No             | $\beta_{CB}$ in M4; $B$ pre |
| <b>S4</b> Joint conditioned            | All three channels together     | Yes                | No (equal)                  | No             | M5 jointly                  |
| <b>S5</b> G10-aware ATM                | Apply ATM where the evidence is | Yes                | No (equal)                  | No             | G10-vs-EM differ            |
| <b>S6</b> Risk scaling, fixed-gross    | Relative de-risking             | No                 | Yes                         | No             | ATM and $B$ predi           |
| <b>S6</b> Risk scaling, variable-gross | Absolute de-risking             | No                 | Yes                         | Yes            | Same, on total ex           |
| <b>S7</b> Expanding M5                 | Let the regression rank         | Yes                | No (equal)                  | No             | Full M5, re-estima<br>ple   |

## HOW TO READ IT, AND COVERAGE

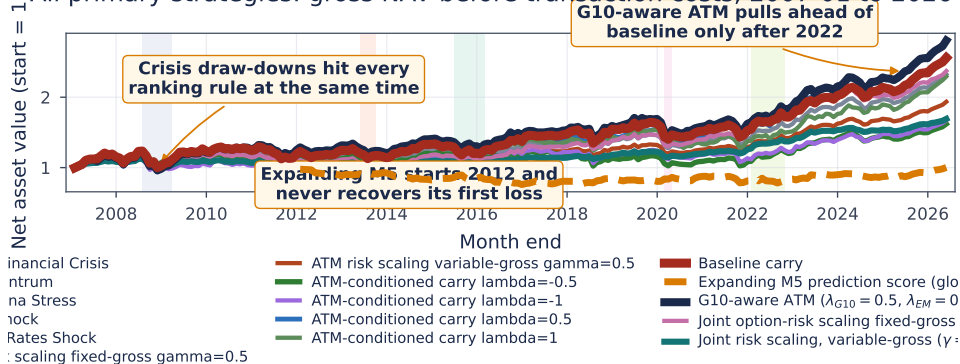
“Changes selection” is true exactly when the rule alters  $Score_{i,t}$  before the sort; “changes gross” only for variable-gross. With the parameter grids these nine rows expand to the **33 strategies** tested: **11 primary** and **22 exploratory**, labelled before any results were seen.



# Gross NAV: all primary strategies before transaction costs

*Crisis windows shaded; costs excluded so the ranking signal is visible on its own*

All primary strategies: gross NAV before transaction costs, 2007-01 to 2026-



## WHAT TO NOTICE

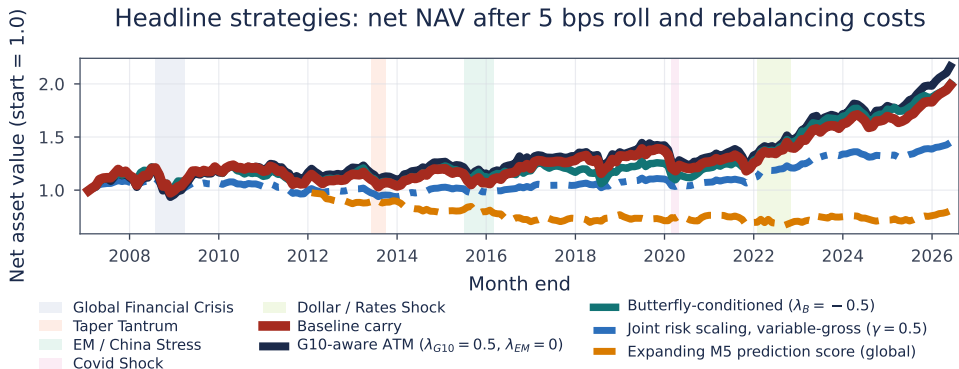
Every ranking rule draws down **together** in the shaded windows — the option conditioners give no crisis diversification. Dispersion opens up in calm periods instead.

## WHY START WITH GROSS

Gross NAV isolates the **information content** of each score. The conditioned rules turn over 30–180% more than baseline, so several look better here than they will after costs.

# Net NAV after 5 bps costs — the main result

Baseline plus the four representative alternatives



## WHAT EACH LINE SAYS

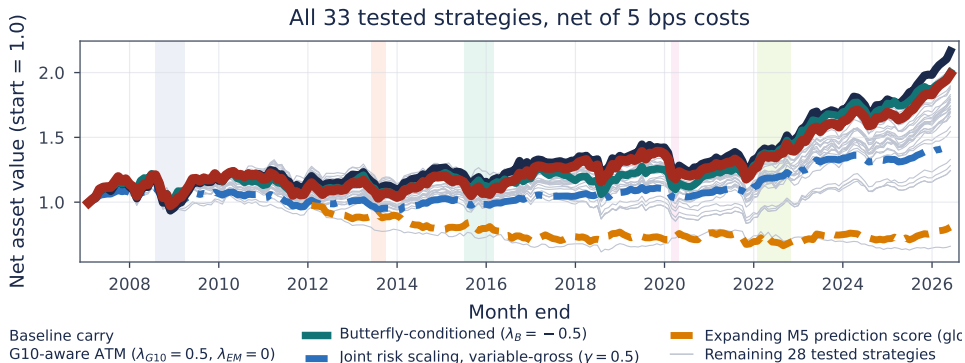
G10-aware ATM improves the endpoint **modestly** (2.17 vs 1.99) and only after 2022. Butterfly conditioning gives a **smoother path** at the same endpoint. Variable-gross **sacrifices upside**. Expanding M5 **fails materially**, ending below 1.0.

## HOW TO READ IT FAIRLY

Expanding M5 starts in **2012**, so its endpoint needs the common 173-month window (appendix; it does not rescue it). And the navy margin over baseline is inside the noise: HAC  $p = 0.317$ .

# The complete strategy grid: every tested rule, net of 5 bps

33 strategies; baseline and the four representatives highlighted, all others in grey



## WHY SHOW ALL 33

Presenting only the winners would be a selection illusion. The grey band is the **full opportunity set** searched: four  $\lambda$  values per conditioner, seven joint combinations, four risk-scaling variants in two gross modes, and two expanding-M5 versions.

## THE HONEST READING

The red baseline sits **comfortably inside** the band, not below it, and roughly half the grid ends above it — what a search over uninformative parameters would produce. Judge the winner's margin against **the dispersion of the whole band**.

# Performance, net of 5 bps costs

Baseline first;  $\Delta$  Sharpe and HAC  $p$ -values are paired against baseline over matched months

| Strategy                                    | Months | Ann. return | Ann. vol. | Sharpe | Max DD  | Turnover | $\Delta$ Sharpe | HAC $p$ |
|---------------------------------------------|--------|-------------|-----------|--------|---------|----------|-----------------|---------|
| Baseline carry                              | 233    | +3.96%      | 9.09%     | +0.44  | -19.40% | 0.18     | —               | —       |
| G10-aware ATM 0.5 / 0                       | 233    | +4.39%      | 8.95%     | +0.49  | -21.20% | 0.23     | +0.055          | 0.317   |
| Butterfly-conditioned lambda=-0.5           | 233    | +3.91%      | 8.01%     | +0.49  | -17.44% | 0.30     | +0.052          | 0.953   |
| Joint risk scaling variable-gross gamma=0.5 | 233    | +2.01%      | 4.62%     | +0.44  | -15.67% | 0.18     | +0.000          | 0.083   |
| Expanding M5 global                         | 173    | -1.26%      | 6.76%     | -0.19  | -33.60% | 0.51     | -0.713          | 0.029   |

## HOW TO READ THE LAST TWO COLUMNS

$\Delta$  Sharpe uses the **months both strategies share**, which is why the expanding-M5 figure ( $-0.713$ ) exceeds the gap in full-sample Sharpe ratios. The HAC  $p$ -value tests whether the **mean monthly return difference** is zero, with three Newey–West lags. Turnover is part of the result — the rules that trade most perform worst.

## NO STRATEGY DOMINATES BASELINE IN RETURN, SHARPE AND DRAWDOWN AT ONCE

G10-aware ATM wins on return (+4.39%) and Sharpe (0.49) but its drawdown is **worse** ( $-21.2\%$  vs  $-19.4\%$ ). Butterfly conditioning matches that Sharpe with a better drawdown but no extra return. Joint variable-gross has the best drawdown and the *same* Sharpe as baseline, by halving return and volatility together.

## DO NOT OVER-READ THE SHARPE GAPS

The best improvement is +0.055 Sharpe with HAC  $p = 0.317$  — **not** distinguishable from zero. The only differences clearing significance are the **failures**: expanding M5 ( $p = 0.029$ ) and the negative- $\lambda$  ATM variants.

# Crisis-period behaviour

Five stress windows fixed in advance in Notebook 07, net of 5 bps; average gross over the same months

## PERIOD RETURN

| Window           | S0      | S5      | S3      | S6 VG  |
|------------------|---------|---------|---------|--------|
| GFC 2008–09      | -10.14% | -11.24% | -10.43% | -7.79% |
| Taper 2013       | -5.34%  | -5.34%  | -6.44%  | -4.14% |
| EM–China 2015–16 | -5.54%  | -4.65%  | -2.73%  | -2.66% |
| Covid 2020       | -7.47%  | -7.12%  | -6.62%  | -3.46% |
| Dollar 2022      | +11.45% | +11.08% | +12.46% | +8.31% |

## MAXIMUM DRAWDOWN

| Window           | S0      | S5      | S3      | S6 VG   |
|------------------|---------|---------|---------|---------|
| GFC 2008–09      | -19.40% | -21.20% | -17.44% | -11.72% |
| Taper 2013       | -8.72%  | -8.72%  | -8.87%  | -5.67%  |
| EM–China 2015–16 | -9.96%  | -8.50%  | -6.14%  | -4.13%  |
| Covid 2020       | -11.27% | -11.15% | -10.45% | -5.13%  |
| Dollar 2022      | -3.30%  | -2.71%  | -2.32%  | -1.67%  |

## WORST MONTH

| Window           | S0     | S5     | S3     | S6 VG  |
|------------------|--------|--------|--------|--------|
| GFC 2008–09      | -9.22% | -9.23% | -9.49% | -4.97% |
| Taper 2013       | -4.62% | -4.62% | -4.65% | -2.61% |
| EM–China 2015–16 | -5.46% | -3.90% | -3.42% | -2.75% |
| Covid 2020       | -9.31% | -8.52% | -8.46% | -4.55% |
| Dollar 2022      | -3.30% | -2.71% | -2.32% | -1.67% |

## AVERAGE GROSS EXPOSURE

| Window           | S0   | S5   | S3   | S6 VG |
|------------------|------|------|------|-------|
| GFC 2008–09      | 2.00 | 2.00 | 2.00 | 1.10  |
| Taper 2013       | 2.00 | 2.00 | 2.00 | 1.09  |
| EM–China 2015–16 | 2.00 | 2.00 | 2.00 | 1.10  |
| Covid 2020       | 2.00 | 2.00 | 2.00 | 1.08  |
| Dollar 2022      | 2.00 | 2.00 | 2.00 | 1.21  |

S0 baseline, S5 G10-aware ATM 0.5/0, S3 butterfly-conditioned  $\lambda_B = -0.5$ , S6 VG joint risk scaling variable-gross  $\gamma = 0.5$ . Expanding M5 is absent: its window starts in 2012. **What to**

**notice:** S6 VG loses least in four of five windows and has the mildest worst month everywhere; S3 helps in EM–China stress but hurts in the Taper Tantrum; S5 is worse than baseline in the GFC.

### THE MECHANISM, STATED PLAINLY

Variable-gross scaling is defensive, but its reduced losses are accompanied by lower average exposure and lower long-run return. It is not forecasting crises, it is holding less — and it gives up upside in the one window where carry made money.

# Strategy takeaways

*What Notebook 07 supports in portfolio tests — kept separate from the regression conclusions*

## 1. G10-AWARE ATM HAS THE HIGHEST TESTED NET SHARPE — MODESTLY

Sharpe 0.49 against baseline's 0.44; return +4.39% against +3.96%. But  $\Delta$  Sharpe is only +0.055 with HAC  $p = 0.317$ , and **drawdown does not improve** (−21.2% vs −19.4%).

## 2. BUTTERFLY CONDITIONING BEHAVES LIKE A FILTER, NOT ALPHA

Return unchanged (3.91% vs 3.96%) while volatility falls from 9.09% to 8.01% and drawdown from −19.4% to −17.4%. It removes risk rather than adding return — consistent with the **carry-quality** reading of  $\beta_{CB}$ , not with a return signal.

## 3. VARIABLE-GROSS RISK SCALING WORKS BY HOLDING LESS

Volatility 4.62% and drawdown −15.7%, both best in class, but return falls to 2.01% and **Sharpe is unchanged** at 0.44. Average gross 1.16 against 2.00: de-risking, not skill.

## 4. EXPANDING M5 FAILS TO TRANSLATE THE REGRESSION INTO A STRATEGY

Return −1.26%, Sharpe −0.19, drawdown −33.6%, turnover 0.51,  $\Delta$  Sharpe −0.71 with HAC  $p = 0.029$  — one of the few differences in the whole study that *is* significant, and it is a loss. The asset-interaction variant is worse (−2.63%), consistent with the coefficient instability shown earlier.

## 5. NO TESTED STRATEGY CLEARLY DOMINATES BASELINE ON ALL DIMENSIONS

Across all **33** strategies, not one beats baseline on annualised return, Sharpe *and* maximum drawdown simultaneously. The option surface, as used here, **redistributes** risk and return rather than expanding the frontier.

## WHAT THIS DOES AND DOES NOT SETTLE

It does **not** show the option surface is uninformative — the magnitude and severe-loss regressions are strong. It shows that **a monthly cross-sectional carry ranking is not the right vehicle** for that information.

# Proposed next steps — not current results

*Nothing on this slide has been tested yet*

## 1. ASYMMETRIC LONG-LEG OVERLAYS

The long leg is 98.7% EM and carries the tail risk; the short leg is 78.6% G10. **Proposed:** apply the risk multiplier to the long leg only, instead of symmetrically.

## 2. REGIME-TRIGGERED GROSS SCALING

Continuous scaling paid for protection in every month. **Proposed:** de-risk only when a signal enters an upper quantile of its own history, leaving normal months untouched.

## 3. SAME-GROSS AND VOLATILITY-TARGET BENCHMARKS

Variable-gross cannot fairly be compared to baseline at gross  
2. **Proposed:** benchmark against a scaled-down baseline and a volatility-targeted baseline, separating de-risking from skill.

## 4. A FROZEN HOLDOUT PERIOD

33 strategies were evaluated on one sample. **Proposed:** freeze the most recent years, finalise the rule on the earlier period only, and report the holdout once.

## 5. BROADER FUNDAMENTAL OR MACRO CONDITIONING VARIABLES

**Proposed:** reuse this framework — pooled ranking, multiplicative clipped conditioner, dual cost model — with macro conditioners in place of option signals. The framework is the transferable contribution.

## 6. FULL-UNIVERSE OPTION-COVERAGE ROBUSTNESS

Six carry currencies lack option-signal coverage and CNH is held out. **Proposed:** quantify how much of the comparison is driven by the 20-currency restriction itself.

## FRAMING FOR THE DISCUSSION

What this week delivers is a **validated, reusable framework** plus a set of **honest negative results** — not a new alpha source. Both are prerequisites for the items above.

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# Appendix

Full tables, alternative inference, cost ladder and walk-forward evidence

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# Appendix A — All 33 strategies, net of 5 bps, ranked by Sharpe

Baseline highlighted

| Strategy                                                  | Mo. | Ann. ret. | Ann. vol. | Sharpe | Max DD  | Turn. | Δ Sharpe | HAC $p$ |
|-----------------------------------------------------------|-----|-----------|-----------|--------|---------|-------|----------|---------|
| G10-aware ATM conditioning g10=0.5 em=0                   | 233 | +4.39%    | 8.95%     | +0.49  | -21.20% | 0.23  | +0.055   | 0.317   |
| Joint option-conditioned carry atm=0 skew=0 bf=-0.5       | 233 | +3.91%    | 8.01%     | +0.49  | -17.44% | 0.30  | +0.052   | 0.953   |
| Butterfly-conditioned carry lambda=-0.5                   | 233 | +3.91%    | 8.01%     | +0.49  | -17.44% | 0.30  | +0.052   | 0.953   |
| Skew risk scaling variable-gross gamma=0.5                | 233 | +3.07%    | 6.48%     | +0.47  | -15.81% | 0.19  | +0.039   | 0.197   |
| Skew risk scaling fixed-gross gamma=0.5                   | 233 | +4.05%    | 8.60%     | +0.47  | -19.79% | 0.24  | +0.036   | 0.809   |
| Butterfly risk scaling variable-gross gamma=0.5           | 233 | +3.03%    | 6.67%     | +0.45  | -16.60% | 0.21  | +0.018   | 0.144   |
| Joint option-risk scaling variable-gross gamma=0.5        | 233 | +2.01%    | 4.62%     | +0.44  | -15.67% | 0.18  | +0.000   | 0.083   |
| Baseline carry                                            | 233 | +3.96%    | 9.09%     | +0.44  | -19.40% | 0.18  | —        | —       |
| Butterfly risk scaling fixed-gross gamma=0.5              | 233 | +3.67%    | 8.53%     | +0.43  | -20.90% | 0.26  | -0.005   | 0.484   |
| ATM-conditioned carry lambda=0.5                          | 233 | +4.03%    | 9.39%     | +0.43  | -18.82% | 0.24  | -0.006   | 0.902   |
| Skew-conditioned carry lambda=-1                          | 233 | +3.45%    | 8.19%     | +0.42  | -17.42% | 0.30  | -0.014   | 0.584   |
| Joint option-risk scaling fixed-gross gamma=0.5           | 233 | +3.41%    | 8.18%     | +0.42  | -24.57% | 0.30  | -0.019   | 0.349   |
| Skew-conditioned carry lambda=0.5                         | 233 | +3.62%    | 8.77%     | +0.41  | -16.39% | 0.25  | -0.022   | 0.649   |
| Butterfly-conditioned carry lambda=0.5                    | 233 | +3.72%    | 9.39%     | +0.40  | -19.24% | 0.26  | -0.039   | 0.703   |
| G10-aware ATM conditioning g10=0.5 em=-0.5                | 233 | +3.19%    | 8.13%     | +0.39  | -18.54% | 0.30  | -0.043   | 0.398   |
| ATM risk scaling fixed-gross gamma=0.5                    | 233 | +3.38%    | 8.70%     | +0.39  | -20.78% | 0.25  | -0.047   | 0.065   |
| ATM risk scaling variable-gross gamma=0.5                 | 233 | +2.59%    | 6.73%     | +0.39  | -17.51% | 0.20  | -0.050   | 0.024   |
| Butterfly-conditioned carry lambda=-1                     | 233 | +2.93%    | 7.76%     | +0.38  | -19.90% | 0.34  | -0.059   | 0.345   |
| Skew-conditioned carry lambda=-0.5                        | 233 | +3.09%    | 8.45%     | +0.37  | -19.17% | 0.29  | -0.070   | 0.262   |
| Joint option-conditioned carry atm=0 skew=-0.5 bf=0       | 233 | +3.09%    | 8.45%     | +0.37  | -19.17% | 0.29  | -0.070   | 0.262   |
| G10-aware ATM conditioning g10=0 em=0.5                   | 232 | +3.52%    | 9.77%     | +0.36  | -19.35% | 0.19  | -0.076   | 0.376   |
| ATM-conditioned carry lambda=1                            | 233 | +3.40%    | 9.49%     | +0.36  | -21.01% | 0.23  | -0.078   | 0.435   |
| Butterfly-conditioned carry lambda=1                      | 233 | +3.47%    | 9.72%     | +0.36  | -24.01% | 0.28  | -0.078   | 0.520   |
| Joint option-conditioned carry atm=0.5 skew=0.5 bf=0.5    | 233 | +3.52%    | 9.89%     | +0.36  | -23.98% | 0.21  | -0.079   | 0.597   |
| Joint option-conditioned carry atm=0.5 skew=-0.5 bf=0.5   | 233 | +3.26%    | 9.72%     | +0.34  | -21.32% | 0.27  | -0.100   | 0.294   |
| Skew-conditioned carry lambda=1                           | 233 | +3.14%    | 9.37%     | +0.34  | -18.05% | 0.25  | -0.100   | 0.339   |
| Joint option-conditioned carry atm=-0.5 skew=-0.5 bf=-0.5 | 233 | +2.56%    | 7.88%     | +0.32  | -18.36% | 0.30  | -0.111   | 0.188   |
| G10-aware ATM conditioning g10=0 em=-0.5                  | 233 | +1.83%    | 8.53%     | +0.22  | -30.51% | 0.29  | -0.220   | 0.005   |
| ATM-conditioned carry lambda=-1                           | 233 | +1.61%    | 8.11%     | +0.20  | -21.39% | 0.29  | -0.237   | 0.009   |
| ATM-conditioned carry lambda=-0.5                         | 233 | +1.47%    | 8.36%     | +0.18  | -31.71% | 0.28  | -0.259   | 0.005   |
| Joint option-conditioned carry atm=-0.5 skew=0 bf=0       | 233 | +1.47%    | 8.36%     | +0.18  | -31.71% | 0.28  | -0.259   | 0.005   |
| Expanding M5 prediction score global                      | 173 | -1.26%    | 6.76%     | -0.19  | -33.60% | 0.51  | -0.713   | 0.029   |
| Expanding M5 prediction score asset-interaction           | 173 | -2.63%    | 7.01%     | -0.38  | -38.43% | 0.59  | -0.903   | 0.007   |

Expanding-M5 rows cover 173 months, all others 233. Δ Sharpe and HAC  $p$  are paired against baseline on matched months, three Newey–West lags.

## Appendix B — Full inference detail for the M5 interactions

Every sample, both standard-error methods

| Coef.        | Sample | Estimate | IID SE  | IID $t$ | IID $p$ | 2-way $t$ | 2-way $p$ | Two-way status        |
|--------------|--------|----------|---------|---------|---------|-----------|-----------|-----------------------|
| $\beta_{CV}$ | All    | +0.00107 | 0.00068 | +1.56   | 0.118   | +1.23     | 0.235     | available             |
| $\beta_{CV}$ | G10    | +0.00277 | 0.00101 | +2.74   | 0.006   | +2.87     | 0.021     | available             |
| $\beta_{CV}$ | EM     | -0.00078 | 0.00104 | -0.75   | 0.451   | —         | n/a       | invalid cov. diagonal |
| $\beta_{CD}$ | All    | -0.00051 | 0.00063 | -0.81   | 0.417   | -1.26     | 0.224     | available             |
| $\beta_{CD}$ | G10    | +0.00040 | 0.00097 | +0.42   | 0.677   | +0.82     | 0.436     | available             |
| $\beta_{CD}$ | EM     | +0.00033 | 0.00141 | +0.24   | 0.813   | —         | n/a       | invalid cov. diagonal |
| $\beta_{CB}$ | All    | +0.00084 | 0.00058 | +1.45   | 0.148   | +1.95     | 0.067     | available             |
| $\beta_{CB}$ | G10    | +0.00085 | 0.00087 | +0.97   | 0.332   | +1.21     | 0.260     | available             |
| $\beta_{CB}$ | EM     | +0.00064 | 0.00114 | +0.57   | 0.572   | —         | n/a       | invalid cov. diagonal |

### READING THE STATUS COLUMN

“invalid cov. diagonal” means the two-way clustered covariance matrix produced a negative estimated variance and was rejected, so only the IID column is available for that cell. This affects the entire EM subsample.

# Appendix C — M1 main effects, full inference

*The result that rules out option signals as unconditional return predictors*

| Coef.     | Sample | Estimate | IID SE  | IID $t$ | IID $p$ | 2-way $t$ | 2-way $p$ | Two-way status        |
|-----------|--------|----------|---------|---------|---------|-----------|-----------|-----------------------|
| $\beta_C$ | All    | +0.00100 | 0.00072 | +1.38   | 0.168   | +3.05     | 0.007     | available             |
| $\beta_C$ | G10    | +0.00082 | 0.00102 | +0.80   | 0.423   | +1.10     | 0.302     | available             |
| $\beta_C$ | EM     | +0.00120 | 0.00102 | +1.18   | 0.238   | —         | n/a       | invalid cov. diagonal |
| $\beta_V$ | All    | +0.00053 | 0.00068 | +0.77   | 0.441   | +0.98     | 0.338     | available             |
| $\beta_V$ | G10    | +0.00004 | 0.00089 | +0.05   | 0.960   | +0.07     | 0.945     | available             |
| $\beta_V$ | EM     | +0.00104 | 0.00107 | +0.98   | 0.328   | —         | n/a       | invalid cov. diagonal |
| $\beta_D$ | All    | -0.00023 | 0.00079 | -0.29   | 0.773   | -0.42     | 0.680     | available             |
| $\beta_D$ | G10    | -0.00002 | 0.00105 | -0.02   | 0.987   | -0.02     | 0.983     | available             |
| $\beta_D$ | EM     | -0.00045 | 0.00149 | -0.30   | 0.761   | —         | n/a       | invalid cov. diagonal |
| $\beta_B$ | All    | -0.00047 | 0.00061 | -0.77   | 0.444   | -1.27     | 0.219     | available             |
| $\beta_B$ | G10    | -0.00022 | 0.00082 | -0.27   | 0.785   | -0.37     | 0.718     | available             |
| $\beta_B$ | EM     | -0.00076 | 0.00116 | -0.65   | 0.515   | —         | n/a       | invalid cov. diagonal |

## Appendix D — All nine G10-versus-EM difference tests

*Pooled M5 with an EM dummy interacted with every regressor*

| Estimate        |                            | Value    | IID $t$ | IID $p$ | 2-way $t$ | 2-way $p$ | 2-way 95% CI         |
|-----------------|----------------------------|----------|---------|---------|-----------|-----------|----------------------|
| G10             | $\beta_{CV}$               | +0.00277 | +2.57   | 0.010   | +2.94     | 0.008     | [+0.00079, +0.00474] |
| EM              | $\beta_{CV} + \delta_{CV}$ | -0.00078 | -0.79   | 0.429   | -0.81     | 0.430     | [-0.00281, +0.00125] |
| <b>EM – G10</b> | $\delta_{CV}$              | -0.00355 | -2.43   | 0.015   | -2.29     | 0.034     | [-0.00680, -0.00030] |
| G10             | $\beta_{CD}$               | +0.00040 | +0.39   | 0.696   | +0.88     | 0.392     | [-0.00056, +0.00137] |
| EM              | $\beta_{CD} + \delta_{CD}$ | +0.00033 | +0.25   | 0.804   | +0.45     | 0.659     | [-0.00122, +0.00189] |
| <b>EM – G10</b> | $\delta_{CD}$              | -0.00007 | -0.04   | 0.967   | -0.09     | 0.930     | [-0.00173, +0.00159] |
| G10             | $\beta_{CB}$               | +0.00085 | +0.91   | 0.364   | +1.23     | 0.235     | [-0.00060, +0.00229] |
| EM              | $\beta_{CB} + \delta_{CB}$ | +0.00064 | +0.59   | 0.553   | +0.94     | 0.362     | [-0.00080, +0.00208] |
| <b>EM – G10</b> | $\delta_{CB}$              | -0.00020 | -0.14   | 0.887   | -0.34     | 0.740     | [-0.00147, +0.00106] |

### ONLY THE ATM CHANNEL DIFFERS ACROSS CLASSES

The carry×ATM difference is  $-0.00355$  with clustered  $p = 0.034$ . The skew and butterfly differences are  $-0.00007$  ( $p = 0.930$ ) and  $-0.00020$  ( $p = 0.740$ ) — no evidence of class heterogeneity in either.

## Appendix E — Fixed-effect robustness for carry × ATM

*Does the interaction survive absorbing currency and month effects?*

| Fixed effects  | Sample | $\beta_{CV}$ | IID $p$ | 2-way $p$ | $N$   | Adj. $R^2$ |
|----------------|--------|--------------|---------|-----------|-------|------------|
| currency       | All    | +0.00088     | 0.238   | —         | 4,658 | -0.0007    |
| currency       | EM     | -0.00134     | 0.225   | —         | 2,561 | -0.0019    |
| currency       | G10    | +0.00385     | <0.001  | —         | 2,097 | 0.0036     |
| currency_month | All    | +0.00081     | 0.146   | —         | 4,658 | 0.4942     |
| currency_month | EM     | -0.00071     | 0.411   | —         | 2,561 | 0.4551     |
| currency_month | G10    | +0.00190     | 0.020   | —         | 2,097 | 0.5839     |
| month          | All    | +0.00105     | 0.039   | —         | 4,658 | 0.4941     |
| month          | EM     | -0.00012     | 0.884   | —         | 2,561 | 0.4546     |
| month          | G10    | +0.00130     | 0.062   | —         | 2,097 | 0.5847     |
| none           | All    | +0.00107     | 0.118   | 0.235     | 4,658 | 0.0011     |
| none           | EM     | -0.00078     | 0.451   | —         | 2,561 | -0.0009    |
| none           | G10    | +0.00277     | 0.006   | 0.021     | 2,097 | 0.0053     |

### WHY THIS CHECK MATTERS

Currency fixed effects absorb permanent level differences between currencies; month fixed effects absorb the common dollar factor. If  $\beta_{CV}$  survived only without them, the result could be an artefact of cross-currency level differences rather than genuine within-currency conditioning.

# Appendix F — Cost sensitivity and the walk-forward window

*Net Sharpe across the full cost ladder, and the common-window comparison*

## NET SHARPE AT EACH COST LEVEL

| Strategy                                       | 1 bp  | 2 bps | 5 bps | 10 bps | 25 bps |
|------------------------------------------------|-------|-------|-------|--------|--------|
| Baseline carry                                 | +0.55 | +0.52 | +0.44 | +0.29  | -0.14  |
| G10-aware ATM 0.5 / 0                          | +0.61 | +0.58 | +0.49 | +0.34  | -0.11  |
| Butterfly-conditioned $\lambda=-0.5$           | +0.63 | +0.59 | +0.49 | +0.32  | -0.20  |
| Joint risk scaling variable-gross $\gamma=0.5$ | +0.57 | +0.54 | +0.44 | +0.26  | -0.26  |
| Expanding M5 global                            | -0.01 | -0.05 | -0.19 | -0.41  | -1.08  |

The base case, 5 bps, is the column behind every headline number in this deck.

## WALK-FORWARD WINDOW: 2012-01 TO 2026-05, 173 MONTHS

| Strategy                                       | Months | Ann. return | Ann. vol. | S |
|------------------------------------------------|--------|-------------|-----------|---|
| Baseline carry                                 | 173    | +4.37%      | 8.30%     |   |
| G10-aware ATM 0.5 / 0                          | 173    | +4.62%      | 7.93%     |   |
| Joint risk scaling variable-gross $\gamma=0.5$ | 173    | +2.66%      | 4.18%     |   |
| Expanding M5 global                            | 173    | -1.26%      | 6.76%     |   |

## TWO THINGS THIS SETTLES

First, the ranking of strategies is **stable across cost assumptions** — no conclusion here depends on choosing 5 bps. Second, restricting to the walk-forward window **does not rescue** the expanding M5 score; it underperforms baseline there too. Exploratory variants are absent from the walk-forward table because Notebook 07 restricts it to strategies flagged primary.

# Appendix G — Risk outcomes by asset class

Full three-sample version of the magnitude and severe-loss regressions

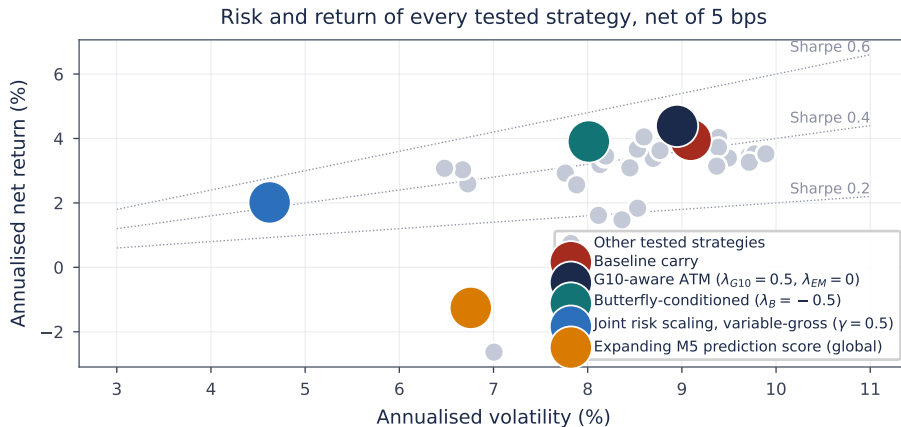
| Left-hand side                | Sample | ATM volatility $V$ |                     | Butterfly $B$ |                    | $R^2$  |
|-------------------------------|--------|--------------------|---------------------|---------------|--------------------|--------|
|                               |        | Coef.              | $p$                 | Coef.         | $p$                |        |
| $E[r_{t+1}]$                  | All    | +0.00081           | 0.080               | -0.00044      | 0.360              | 0.0026 |
|                               | G10    | +0.00016           | 0.537               | -0.00046      | 0.539              | 0.0086 |
|                               | EM     | +0.00172           | 0.157 <sup>†</sup>  | -0.00114      | 0.351 <sup>†</sup> | 0.0018 |
| $E[ r_{t+1} ]$                | All    | +0.00516           | <0.001              | +0.00099      | 0.083              | 0.0742 |
|                               | G10    | +0.00277           | <0.001 <sup>†</sup> | +0.00061      | 0.280 <sup>†</sup> | 0.0303 |
|                               | EM     | +0.00821           | <0.001              | +0.00129      | 0.171              | 0.1167 |
| $E[r_{t+1}^2]$                | All    | +0.00039           | <0.001              | +0.00014      | 0.023              | 0.0402 |
|                               | G10    | +0.00019           | <0.001 <sup>†</sup> | +0.00009      | 0.072 <sup>†</sup> | 0.0267 |
|                               | EM     | +0.00063           | 0.001               | +0.00017      | 0.106              | 0.0500 |
| $P(\text{severe loss}_{t+1})$ | All    | +0.03322           | <0.001              | +0.01342      | 0.068              | 0.0187 |
|                               | G10    | +0.02216           | 0.003               | +0.01725      | 0.145              | 0.0130 |
|                               | EM     | +0.04566           | 0.001               | +0.00927      | 0.433              | 0.0270 |

## READING THE DAGGER

<sup>†</sup> marks a cell where the two-way clustered covariance matrix was rejected and the IID  $p$ -value is quoted as a flagged benchmark. ATM volatility predicts magnitude and severe loss in **both** classes; the effect is roughly three times larger in EM, where  $R^2$  on  $|r_{t+1}|$  reaches 11.7%.

# Appendix H — Risk and return of the full strategy grid

Every tested strategy against constant-Sharpe rays



The grid spans volatility from about 4.6% to 9.9%, but Sharpe ratios cluster tightly between 0.32 and 0.49 apart from the two expanding-M5 failures and the negative- $\lambda$  ATM variants. Most of the visible variation is *exposure*, not skill.